

MODULE 8

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Assignments : Introduction to CSS

Q1] What is CSS? Explain why CSS is required in web development.

→ CSS (Cascading style sheets) is a stylesheet language used to control the presentation, layout, and visual appearance of HTML web pages. It defines how HTML elements should be displayed on different devices and screen sizes.

Why CSS is required :

- Separates content (HTML) from design (CSS)
- Improves website appearance (colors, fonts, layouts).
- Makes websites responsive for mobile and desktop.
- Reduces code repetition and improves maintainability.
- Enhances user experience and readability.
- Helps in SEO and accessibility.

Without CSS, websites would look plain and unattractive.

2] Write the differences between HTML and CSS.

HTML	CSS
① HTML stands for Hypertext Markup Language.	① CSS stands for Cascading Style Sheets
② Used to create structure and content.	② Used to style and design content.
③ Defines elements like text, images, forms	③ Defines colors, fonts, layout
④ Uses tags	④ Uses selectors and properties
⑤ Content-focused	⑤ Design-focused.

3] Explain the basic syntax of CSS with an example.

Basic CSS - Syntax ✓

```

Selector {
    property : value;
}

```

Example :

```
p {
    color: blue;
    font-size: 16px;
}
```

- p → selector
- color, font-size → properties
- blue and 16px → values

This CSS styles all paragraph elements.

Q1 What are the different types of CSS?
Explain Inline, Internal, and External CSS with advantages and disadvantages.

→
1) Inline CSS :-

- Applied directly inside an HTML element.
- example :

```
<p style="color:red">Hello</p>
```

Advantages :-

- 1) Quick styling
- 2) Useful for testing

Disadvantages :-

- 1) Breaks separation of concerns
- 2) Difficult to maintain
- 3) Not reusable

2) Internal CSS :-

- written inside <style> tag in <head>
- example :-

<style>

p {

color: red;

}

</style>

Advantages :-

- ① useful for single-page styling
- ② No external file needed.

Disadvantages :-

- ① Not reusable
- ② Increases page size

3) External CSS :-

- written in a separate .css file
- example :-

<link rel="stylesheet" href="style.css">

Advantages :-

- ① Reusable across pages
- ② Easy maintenance
- ③ Faster page loading (cached)

Disadvantages :-

- Requires extra file.

Q] Write a note on CSS comments. Why are comments important in CSS?

→ CSS comments are used to explain code and are ignored by browsers.

Syntax :-

`/* This is a CSS comment */`

Importance :-

- ① Improves code readability
- ② Helps team collaboration
- ③ Useful for debugging
- ④ Explains complex styles
- ⑤ Helps during future updates.

Q] Explain the terms :

- 1) CSS selector :
- 2) CSS property
- 3) CSS value

→ CSS Selector:

Used to select HTML elements to style

Example: p, .class, #id

CSS property:

Defines what to style

Example: color, font-size

CSS Value:

Specifies how to style

Example: red, 20px

7 Write the priority order of CSS when multiple styles are applied

→ CSS Priority Order (Highest to Lowest):-

1) Inline CSS

2) Internal CSS

3) External CSS

4) Browser default styles

!important → overrides all styles
(not recommended)

8] Why is external CSS preferred in real websites?

→ Reasons : —

- ① Easy maintenance
- ② Reusability across multiple pages.
- ③ Faster page loading due to caching
- ④ Clean HTML code
- ⑤ Better Scalability for large projects.

9] What problems can occur if only inline CSS is used in a large website?

→

- ① code duplication
- ② Difficult maintenance
- ③ Increased page size
- ④ poor readability
- ⑤ Time-consuming updates
- ⑥ Not Scalable
- ⑦ Violates best practices.

"HTML gives structure, CSS gives beauty, and Javascript gives behavior."

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12] Write a short note on real-life use of CSS in websites.

→ CSS is used to :

- ① Design website layouts
- ② Create responsive designs
- ③ Add Animations and transitions
- ④ Improve accessibility
- ⑤ Enhance user experience
- ⑥ Maintain brand identity
- ⑦ Style real-world websites like e-commerce, blogs, and dashboards.

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Assignment : CSS selectors

1] What is a CSS Selector? why are selectors important in CSS?

→ A CSS selector is used to select HTML elements that you want to style. Selectors tell the browser which elements the CSS rules should apply to.

Importance of selectors :

- ① Target specific HTML elements
- ② Apply styles efficiently
- ③ Improve code readability
- ④ Reduce repetition
- ⑤ make CSS scalable and maintainable
- ⑥ Essential for responsive and dynamic designs.

2] Explain element selector with syntax and one example.

→ An element selector selects all instances of a specific HTML tag.

Syntax : -

```
element {  
    property : value;  
}
```

Example :

```
p {
    color: blue;
}
```

This styles all <p> elements with blue text.

③ Explain class selector and ID selector.

→

Class Selector —

- Used to style multiple elements.
- Uses . (dot)

- Example : —

```
.box {
    border: 1px solid black;
}
```

ID Selector:

- Used to style one unique element
- Uses # (hash)

- Example : —

```
#header {
    background-color: lightgray;
}
```


Q] What is a grouping selector?

→ A grouping selector applies the same styles to multiple elements.

Syntax:

```
h1, h2, p {
    color: green;
}
```

Usefulness : —

- ① Reduces duplicate code
- ② Saves time
- ③ Improves readability
- ④ Easier maintenance

Q] Explain nested (descendant) selectors with an example.

→ A nested or descendant selector selects elements that are inside another element.

Syntax:

```
parent child {
    property: value;
}
```

Example :—

```
div p {
    color: red;
}
```

Styles only <p> elements inside <div>.

- Q) What are attribute selectors? Explain any three types of attribute selectors.
- Attribute selectors select elements based on attributes or attribute values.

1) [attribute]

```
input [required] {  
    border: 2px solid red;  
}
```

2) [attribute = "value"]

```
input [type = "text"] {  
    background-color: lightyellow;  
}
```

3) [attribute ^ = "value"] (starts with)

```
a [href ^ = "https"] {  
    color: green;  
}
```

7] What are pseudo-classes; write names of any four pseudo-classes.

→ pseudo-classes define the special state of an element.

Examples : —

- :hover
- :active
- :focus
- :visited

```
a:hover {  
    color: red;  
}
```

8] What are pseudo-elements; write names of any four pseudo-elements.

→ Pseudo-elements style specific parts of an element.

Examples —

- ::before
- ::after
- ::first-letter
- ::first-line

```
p ::first-letter {  
    font-size: 30px;  
}
```


CSS selectors help target elements precisely, making design flexible, scalable, and maintainable.

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9] Write the difference between pseudo-class and pseudo-element.

pseudo-class	pseudo-element
① styles elements states	① Styles parts of elements
② Uses single :	② Uses double ::
③ Example :- :hover	③ Example :- ::before

10] Write a short note on real-life use of CSS selectors in websites.

→ CSS selectors are used to:

- ① Style navigation menus
- ② Design forms and buttons
- ③ Apply hover effects
- ④ Build responsive layouts
- ⑤ Improve accessibility ::
- ⑥ Create interactive UI elements.
- ⑦ maintain large-scale websites efficiently

Assignment : CSS Colors, Units, & Backgrounds

1) Difference between HEX and RGB color formats (with examples).



HEX color format —

- Uses hexadecimal values
- Written as #RRGGBB
- Each pair represents Red, Green, Blue.
- Values range from 00 to FF

Example :—

Color : #ff0000; /* red */

RGB Color Format —

- Uses decimal values
- Written as rgb(red, green, blue)
- Values range from 0 to 255.

Example :—

Color : rgb(255, 0, 0); /* red */

Key difference :—

- HEX uses hexadecimal notation
- RGB uses numeric decimal values

2) Explain RGBA. What does the alpha value represent? What is its range?



RGBA stands for Red, Green, Blue, Alpha

- Alpha represents opacity (transparency).

- Range of alpha: 0 to 1

Alpha value	meaning
0	fully transparent
1	fully opaque

Example 1:-

`background-color: rgba(0,0,0,0.5);`

Creates a semi-transparent black background.

3) HEX code for white and black.



Color	HEX code
white	#FFFFFF
black	#000000

4) How is % Unit Calculated for width?
What does it depend on?

→ The percentage (%) width is calculated relative to the width of the parent element.

Example ! -

```

    • child {
      width : 50%;
    }
  
```

The element takes 50% of its parent's width

Depends on :

- Parent element's width
- Container size

5) Difference between em and rem (with real example)

→ em ! -

- Relative to parent element's font size
- Can cause compounding issues

rem : -

- Relative to root (html) font size
- More predictable

Example ! -

```

    html {
      font-size : 16px;
    }
  
```

```
...-parent {  
    font-size: 20px;  
}
```

```
child-em {  
    font-size: 2em; /* 40px */  
}
```

```
child-rem {  
    font-size: 2rem; /* 32px */  
}
```

Q When should you prefer rem over em in a Scalable UI?

→ Use rem when:

- ① Building large, scalable layouts
- ② Maintaining consistent sizing
- ③ Avoiding nested size calculation issues
- ④ Creating responsive designs

rem ensures uniform sizing across the website.

7) What are vh and vw? Give one practical use case for each.



Vh (Viewport Height)

- 1vh = 1% of Viewport height.

Use case:

```
.hero {
```

```
    height: 100vh;
}
```

Full-screen hero section.

Vw (Viewport width)

- 1vw = 1% of Viewport width

Use case:

```
.container {
```

```
    width: 80vw;
}
```

Responsive container width.

8) Difference between background-size: cover and Contain.



background-size: cover;

background-size: contain;

Cover	Contain
① Fills entire container	① Fits entire image
② Image may be cropped	② Image not cropped
③ Best for banners	③ Best For logos.

9) Explain background-repeat types

1) repeat (default)

- Repeats image both horizontally and vertically.

Example 1 —

`background-repeat : repeat;`

2) no-repeat

- shows image only once

Example : —

`background-repeat : no-repeat;`

"Understanding CSS units and background properties is essential for responsive and visually consistent designs".

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3) repeat - x

- Repeats horizontally only

Example :

```
background-repeat: repeat-x;
```

4) repeat - y

- Repeats vertically only

Example :

```
background-repeat: repeat-y;
```

10] What is background shorthand? Write an example.

→ Background shorthand combines multiple background properties into one line.

Syntax order : —

```
background: image repeat position / size;
```

Example : —

```
background: url("bg.jpg") no-repeat center/cover;
```

Assignment : CSS Text & Font Properties

1] What are text properties in CSS?
Explain why text styling is important in web design?

→ Text properties in CSS are used to control the appearance of text on a webpage; such as color, alignment, Spacing, decoration, and transformation.

Common text Properties :—

- ① color
- ② text-align
- ③ text-decoration
- ④ text-transform
- ⑤ line-height
- ⑥ letter-spacing
- ⑦ word-spacing

Importance of text Styling:—

- ① Improves readability
- ② Enhances User Experience
- ③ Makes content visually attractive
- ④ Helps convey information clearly
- ⑤ Supports branding and consistency.

2] Explain the color property in CSS. Write different ways to define colors.

→ The color property is used to set the text color of an element.

Ways to define colors: -

1] Color name

color: red;

2] HEX value

color: #ff0000;

3] RGB

color: rgb(255, 0, 0);

4] RGBA

color: rgba(255, 0, 0, 0.5);

5] HSL

color: hsl(0, 100%, 50%);

3] Explain text-align property with its values and use cases.

→ The text-align property controls horizontal alignment of text.

Value

Use cases

left



default text alignment

right



Align prices or side lab

center



Headings, banners

justify



Newspapers, blogs

Example :-

```
P {
```

```
text-align: justify;
```

```
}
```

7] What is text-decoration? Explain its values with real-life examples.

→ The text-decoration property adds decorative lines to text.

Value

underline



Example use
links

line-through



Discounted prices

overline



Rarely used

none



Remove link underline

Example :- a {

text-decoration: none;
}

Q Explain text-transform. Why is it useful in headings and titles?

→ The text-transform property controls letter casing.

Values :

- Uppercase
- lowercase
- Capitalize

Example :- h1 {

text-transform: uppercase;
}

Usefulness :-

- ① maintains consistent style
- ② Avoids editing Content text
- ③ Improves heading visibility
- ④ Useful For titles and menus

Q What is line-height? Why is it considered the most important text property?

→ line-height controls the vertical space between lines of text.

Examples —

```
p {
  line-height: 1.6;
}
```

Importance —

- ① Improves readability.
- ② Prevents text clutter.
- ③ Essential for long paragraphs.
- ④ Enhances accessibility.

7) Explain the following font properties

a) Font-family :—

Defines the typeface used.

Example :—

```
body {
  font-family: Arial, sans-serif;
}
```

Use multiple fonts as fallback

b) Font-size :—

Controls the size of text

Example :—

```
p {
  font-size: 16px;
}
```

Units : px, em, rem, %

8] What is font-weight? Explain keyword values and numeric values.

→ Font-weight defines thickness of text

Keyword values:-

- normal
- bold
- lighter
- bolder

Numeric Values :-

- 100 (thin)
- 400 (normal)
- 700 (bold)
- 900 (extra bold)

Example :- $h_1 \Sigma_2$
font-weight : 700;
2

9] What are Google Fonts? Write advantages and disadvantages.

→ Google Fonts are free, web-hosted fonts provided by Google.

Advantages:—

- ① free and easy to use
- ② Large font collection
- ③ Cross-browser support
- ④ No installation required.

"Good Typography improves Usability, accessibility, and brand perception!"

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Disadvantages :-

- ① Requires internet connection
- ② Slight performance impact
- ③ Dependency on external server.

10] Short note on text alignment and spacing in web pages. why should spacing be used carefully?

→ Text alignment and spacing improve content structure and readability.

Importance :-

- ① Guides user attention
- ② Makes content readable
- ③ Improves UI aesthetics
- ④ Supports accessibility

Why Careful Spacing is needed :-

- Too little spacing → cluttered text
- Too much spacing → broken layout
- Balanced spacing improves UX